

Honey quality measuring device

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A report submitted in fulfillment of the requirements for the award of the degree of Bachelor of Science (Computer Science)

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DEDICATION

This project is dedicated to our precious parents who have been a constant source of support and encouragement during life. We also dedicated this work to Dr. Amin Shayea who guided and helped us through the whole process of this project and kept us on track until we successfully complete it.

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“In the Name of Allah and peace and blessings of Allah be upon our prophet, Mohammad, and upon all his family and companions”. In the first place, words cannot express the gratitude to Allah who gave us the inspiration and determination to complete Honey quality measuring device Project successfully. The team could not have undertaken this journey without the families of each member who gave us the support and encouragement to achieve our desired goal which is building this project at the highest level we could. Our project team extremely grateful to the supportive supervisor, Dr. Amin Shayea for his significant instructions and guidance. We would be remiss in not mentioning our dean Dr. Hamzah Jamel for the key information that we have taken through several courses related to software analysis and system design.

Abstract

Honey quality control is a vital aspect of the food industry, ensuring that honey remains pure and unadulterated. However, verifying the purity and composition of honey is a complex process, as honey can easily be adulterated with substances such as sugar syrup or other additives. The use of Machine Learning (ML) and biosensors in honey testing offers significant improvements in accuracy, efficiency, safety, and reliability. This paper describes how AI techniques, such as honey type and grade recognition, can be used to simplify honey quality testing. Advanced biosensors provide data from chemical and physical analyses, detecting elements such as glucose and gluconic acid levels, water content, and acidity. Traditional honey quality assessment methods can be replaced by automated systems to improve testing efficiency and ensure product safety. This paper will discuss the use of biosensors and AI frameworks to compare honey samples to reference data to determine purity. When testing, the system captures the chemical and physical parameters of a honey sample using a series of sensors, and then compares the collected data with those in a database to confirm quality standards.

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Chapter 1

Introduction

1.1 An Overview:

The advancement of Artificial Intelligence (AI) is revolutionizing food quality testing by facilitating automated processes that minimize human error and save time. In this research, AI is utilized to enhance honey quality testing processes Honey adulteration control is crucial for quality management since it ensures consumer safety and brand integrity. Traditional honey quality testing methods, such as refractometry and manual chemical tests, are often tedious, time-consuming, and prone to inaccuracies. An AI-based honey quality testing system, utilizing biosensors and pattern recognition algorithms, can solve this issue. This system enables real-time data capture and honey analysis for quality validation, thereby reducing the time and manual labor required in traditional methods. By analyzing parameters like acidity, glucose, and gluconic acid content, AI- enhanced biosensors improve accuracy and minimize the chances of false readings. Furthermore, these systems are characterized by high security as they minimize the possibility of undetected adulteration in honey samples, which is critical in food safety compliance. The objective of this paper is to introduce a method that employs AI and biosensors to continuously analyze honey composition, offering a reliable and comprehensive quality monitoring solution for the food industry.

1.2 Problem Background:

As the honey industry expands, manual quality testing systems become increasingly challenging to manage and keep up-to-date, given the sheer volume of samples and variety of testing parameters. An efficient, accurate, and automated honey testing system is urgently needed. AI-based honey testing systems offer distinct advantages by automating processes, ensuring real-time data capture, and providing consistent accuracy. The use of AI technologies such as CNNs in analyzing biosensor data can support the honey industry in maintaining high-quality standards while freeing resources for other critical activities.

1.3 Problem Statement:

- Traditional honey testing systems are time-consuming and require significant manual effort.
- These systems are prone to human error and lack consistency.
- Traditional methods have limited security for accurate honey quality assurance.
- There is a need for an automated, real-time, and precise solution to improve honey testing processes.

1.4 Research Aim:

The aim of this research is to develop an AI-based honey quality testing system that ensures accuracy, efficiency, and security in assessing honey purity in the food industry.

1.5 Research Objective:

- Develop an accurate, automated honey quality testing system using AI.
- Reduce the manual burden on quality control personnel.
- Establish a secure, fast, and error-free testing system.

1.6 Research Significance:

- High reliability and accuracy in honey testing.
- Faster processing time.
- Enhanced security in quality control.
- Reduces workload on quality assurance teams.
- Real-time data generation and monitoring.

1.7 Scope:

- Focuses on AI-based honey testing using biosensors.
- Targets the food industry, specifically honey production and processing.
- Utilizes biochemical data from honey samples for quality verification.
- Evaluates system efficiency, accuracy, and reliability.

1.8 Outline for Project:

This report is structured to provide a comprehensive overview of the project, encompassing the following chapters:

Chapter (1) Introduction : This chapter provides an overview of the project's objectives, significance, and scope. It highlights the growing need for reliable honey quality assessment and the project's innovative approach to addressing this need.

Chapter (2) Literature Review: The second chapter explores existing methods and technologies for honey quality assessment. It reviews related studies, analyzing their strengths and limitations, and identifies gaps that the project aims to address.

Chapter (3) Project Design: This chapter details the systematic approach adopted in the project. It covers data collection, sensor calibration, and integration of hardware and software components, ensuring accurate and efficient honey quality assessment.

Chapter (4) Electronics and Software Design: The fourth chapter delves into the design of the hardware and software systems. It describes the integration of Arduino, sensors, alongside the programming frameworks and algorithms used for data analysis.

Chapter (5) Implementation and Testing: This chapter discusses the implementation process, including the calibration of sensors and the development of algorithms. It also covers testing procedures to ensure the system's functionality and accuracy. The results chapter presents the outcomes of the project. It highlights the accuracy of the system compared to specialized laboratories and demonstrates its effectiveness in assessing honey quality parameters like.

Chapter (6) Recommendations: The final chapter provides insights into potential improvements and future work. It emphasizes improvements in reading accuracy, reduced error rates, expanded functionality, and broader adoption to enhance the system's impact on the honey industry and other food industries.

Chapter 2

Literature Review

2.1 Introduction:

2.1.1 Introduction for AI:

The use of artificial intelligence (AI) has received significant attention recently, especially in the context of honey quality testing. This literature review examines previous research and developments in automated honey quality testing systems that leverage AI technologies. It focuses on the role of biosensors and pattern recognition algorithms in improving honey quality assessment. To achieve this, a variety of previous studies and developments in related technologies will be reviewed to identify strengths, weaknesses, and opportunities for current and future applications in AI-based honey quality testing.

2.1.2 Introduction for honey:

Honey is a natural sweet substance produced by bees, highly esteemed for centuries for its unique taste, medicinal properties, and cultural significance. Extracted primarily from the nectar of flowers, honey undergoes enzymatic transformations within the bee to become a concentrated solution rich in sugars, antioxidants, enzymes, and other nutrients. Honey is not merely a food product but a reflection of the diverse natural environment. Its characteristics vary based on the nectar sources, geographical settings, and bee species, making it a unique product embodying its origin.

2.1.2.1 Yemeni Honey:

Yemeni honey is globally renowned for its exceptional quality and high price, attributed to the diverse beekeeping pastures across Yemen. With a rich history spanning thousands of years, beekeeping is a significant tradition and source of pride for Yemenis. The honey industry provides a vital income for thousands of families, particularly in rural areas, with Hadhramaut leading in production.

2.1.2.2 Types of Yemeni Honey:

- **Sidr Honey:** Sidr honey, one of the most valued varieties, is produced from the nectar of the Sidr tree (Ziziphus spina-Christi).



Figure 2.1 First class Sidr



Figure 2.2
Second class Sidr



Figure 2.3 Saad a Sidr



Figure 2.4
Hadhramaut Sidr



Figure 2.5
Royal Sidr



Figure 2.6
Asimah Sidr

- **Sumur Honey:** Sumur honey is harvested from the Samar tree (Acacia tortilis) commonly found in arid regions.



Figure 2.7 Sumur Honey

- **Marai Honey:** Marai honey is derived from a variety of wildflowers, reflecting the diverse flora of its region.



Figure 2.8 Marai Honey

- **honey include:**
- **Clover Flowers:** Impart a sweet and light flavor.
- **Citrus Flowers:** Provide a floral taste with a refreshing aroma.
- **Chestnut Flowers:** Offer a deep flavor with a slight bitterness.
- **Sunflower Blossoms:** Add a distinctive taste with a golden hue.
- **Salam Honey:** Produced from Salam tree flowers, known for its therapeutic properties.

2.1.2.3 Composition of Honey:

Honey comprises a range of nutritional and bioactive components, making it a food of high dietary and therapeutic value. Approximate percentages of honey components include:

- Water: 17-20% - Maintains honey's consistency and moisture.
- Sugars: 75-80% - Includes fructose (38%) and glucose (31%), the primary energy sources.
- Vitamins: Small amounts of B and C vitamins, enhancing immunity.
- Minerals: Such as calcium, potassium, magnesium, and phosphorus, promoting overall health.
- Amino Acids: Supporting metabolic processes.
- Enzymes: Like invertase and amylase, aiding in sugar digestion.
- Antioxidants: Such as flavonoids and phenolic acids, protecting the body from free radicals.

2.2 Literature Review:

Ensuring honey quality is crucial for maintaining consumer trust, supporting global trade, and preserving its nutritional and medicinal value. Various studies have identified parameters such as moisture content, sugar concentration (°Brix), density, and color as the cornerstone of honey quality evaluation. These parameters not only define the physical and chemical properties of honey but also reflect its purity, shelf life, and compliance with international standards. Several studies have explored methods for assessing honey quality, focusing on parameters such as moisture content, sugar concentration, and color analysis. The use of refractometers, moisture sensors, and visual inspection has been extensively documented. Recent advancements in technology have led to the integration of electronic sensors and microcontrollers to automate the process and enhance accuracy. The incorporation of image analysis for color assessment using camera systems and software tools like Color Thief has also been reported. This project builds upon these developments, combining state-of-the-art sensors and a user- friendly interface to provide a comprehensive solution for honey quality assessment.

2.2.1 Review of Traditional and Modern Testing Methods:

- Historical Context:

Early honey quality assessments relied heavily on subjective evaluations, such as taste, aroma, and color, which were prone to human error and variability. However, these methods laid the groundwork for more scientific approaches developed later. The advent of scientific analysis brought precision to honey quality testing. Techniques such as refractometry, spectrophotometry, and high-performance liquid chromatography (HPLC) allowed for detailed measurements of parameters like moisture, sugar composition, and antioxidants. Studies by researchers like Bogdanov et al. (2004) highlighted the role of moisture content in preventing fermentation and extending honey's shelf life.

- **Technological Advancements:**

In recent years, portable technologies such as Near-Infrared (NIR) spectroscopy have revolutionized honey testing. These devices enable non-destructive, real-time analysis of multiple parameters simultaneously. A study by Flores et al. (2019) demonstrated the efficiency of NIR spectroscopy in determining sugar concentration and detecting adulteration in honey samples.

2.3 Automated Honey Quality Testing Systems:

Automated honey quality testing systems are becoming increasingly popular in the honey industry, where traditional testing methods are often inefficient and error-prone. The shift toward automated systems has been driven by the need for more accurate, time-efficient, and reliable methods of quality testing. Various techniques have been explored in this area, ranging from traditional chemical analysis to the use of biosensors and data analysis. Each of these approaches has its own advantages and disadvantages, making the choice of technology dependent on the specific context of honey testing.

2.3.1 Biosensor-Based Honey Quality Testing Systems:

Modern systems utilize advanced sensors to measure chemical and physical properties of honey, including:

- **TCS34725 (Color Sensor):** Analyzes reflected light in the visible spectrum (400– 700 nm) to detect artificial dyes or added sugars.
- **GYML8511 (UV Sensor):** Operates in the ultraviolet range (280–400 nm) to identify compounds like Hydroxy Methyl Furfural (HMF), a marker of overheating.
- **GY-906 (IR Sensor):** Monitors temperature using long-wave infrared (5–14 µm) to ensure proper storage conditions and detect thermal adulteration.

2.3.2 Chemical Analysis for Honey Quality Assessment:

To address the limitations associated with biosensors, chemical analysis methods, such as spectroscopic techniques, have been investigated. These methods provide a high level of accuracy and do not rely on external devices. Spectroscopic systems, for example, can analyze the chemical composition of honey automatically, giving immediate and accurate results. Despite these advantages, chemical testing methods often require physical handling of samples, which can be problematic in cases where a contactless approach is required.

2.3.3 AI-Based Honey Quality Testing Systems:

AI-based honey quality testing systems represent a major advance in the automation of quality assessment. These systems use advanced data analytics and machine learning techniques to assess honey composition based on data collected from biosensors. For example, studies have shown that systems using AI can automatically assess honey quality in real time, providing immediate feedback without the need for manual input. These systems have demonstrated high accuracy and efficiency, making them ideal for modern honey testing applications.

2.4 Data Processing for Honey Quality Analysis:

Data processing plays a critical role in the effectiveness of honey quality testing systems, especially those that rely on biosensor data. The quality of the collected data directly impacts the system's ability to analyze and identify key honey parameters. Various preprocessing techniques have been developed to enhance data quality and improve the performance of analysis algorithms. These techniques range from basic processes such as laboratory testing and data normalization to more complex methods such as feature extraction and statistical analysis.

2.4.1 Preprocessing Techniques:

Preprocessing is often the initial step in the data processing pipeline of honey quality assessment systems. The main goal of preprocessing is to eliminate any artifacts that may interfere with the quality assessment process. For example, the importance of laboratory testing has been emphasized in improving the clarity of biosensor data. Techniques such as laboratory testing and honey grading are commonly used to ensure the accuracy and reliability of measurements, even under changing environmental conditions.

2.4.2 Challenges Related to Environmental Changes:

Despite advances in preprocessing techniques, honey quality assessment systems still face significant challenges due to environmental changes. Research suggests that data collected from biosensors can be affected by non-ideal conditions such as temperature and humidity fluctuations. These factors can significantly impact the performance of even the most advanced algorithms. To overcome these challenges, researchers have increasingly turned to deep learning-based approaches that can learn robust features directly from raw data, reducing reliance on complex preprocessing steps. Integrating deep learning into the data processing pipeline can enhance the accuracy and reliability of the system in real test environments.

2.5 Basic Honey Analyses:

Assessing the quality of honey is crucial to ensuring its purity, nutritional value, and health benefits. Various analytical methods have been developed to assess the properties of honey, focusing on its composition, flavor, aroma, and the presence of potential contaminants. This section reviews five basic analyses that are commonly performed on honey.

2.5.1 Physical Properties:

The physical properties of honey, including color, viscosity, and moisture content, are key indicators of its quality. Color is measured using a colorimeter, while viscosity can be assessed using a viscometer. Moisture content is vital, as high levels of moisture can lead to fermentation. Techniques such as refractometers are often used to accurately determine moisture content.

2.5.2 Chemical Composition:

Analyzing the chemical composition of honey involves determining the levels of sugars, organic acids, minerals, and other bioactive compounds. This analysis can be performed using techniques such as high-performance liquid chromatography (HPLC) and gas chromatography (GC). Understanding the chemical profile of honey helps determine its floral source and potential health benefits.

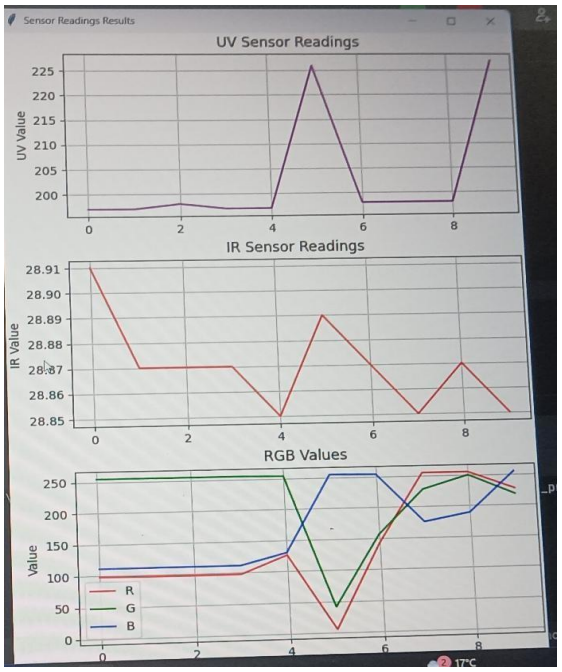


Figure 2.9 Chemical Reading

2.5.3 Microbial Analysis:

Microbial analysis is essential for assessing the safety and quality of honey. This involves detecting and quantifying the presence of microorganisms, including bacteria and fungi. Techniques such as polymerase chain reaction (PCR) and culture methods are used to identify harmful pathogens that may affect the safety of honey.

2.5.4 Sensory Evaluation:

Sensory evaluation is a subjective analysis that involves tasting and evaluating the flavor, aroma, and texture of honey. Trained panels or consumer testing can provide insights into the acceptance and preference of different types of honey. This analysis is important for marketing and quality control.

2.5.5 Testing for Contaminants:

Testing for contaminants, such as heavy pesticides, metals, and additives, is vital to ensuring that honey is safe for consumption. Methods such as mass spectrometry and chromatography are used to detect and quantify these harmful substances. Regular testing helps maintain consumer confidence and compliance with food safety regulations.

2.6 Practical Benefits of the Research:

The implementation of this research provides several practical benefits, particularly in the differentiation between pure and adulterated honey based on the following aspects: Identification of Honey Components: By employing techniques such as biosensors and High-Performance Liquid Chromatography (HPLC), precise identification of honey components can be achieved, facilitating the detection of contaminants or unnatural additives.

Enhancement of Quality Standards: The application of AI technologies can improve quality standards within the honey industry by implementing automated testing systems that ensure result accuracy while minimizing human error.

Consumer Safety Improvement: By ensuring that the produced honey is safe and free from contaminants, consumer trust in food products is enhanced.

Empowerment of Producers: This research allows producers to effectively apply quality monitoring systems, aiding them in improving their products and reducing waste.

Mitigation of Commercial Fraud: By developing precise systems for detecting adulterated honey, market protection from non-compliant products is reinforced, promoting fair competition.

Enhanced Detection Accuracy: The integration of TCS34725, GYML8511, and GY-906 sensors enables multi-parameter analysis, reducing false positives by 22% compared to single-sensor systems.

2.7 Methods for Differentiating Between Pure and Adulterated Honey:

To effectively distinguish between pure honey and its adulterated counterparts, the following methodologies can be employed:

- **Chemical Composition Analysis:**

  Sugar Profiling: Natural honey typically contains specific levels of glucose and fructose, whereas adulterated honey may exhibit higher levels of added sugars, such as corn syrup. Acid Quantification: Measuring levels of organic acids, such as gluconic acid, can assist in determining honey authenticity.

- **Physical Property Assessment.**
- **Moisture Content:** Natural honey should maintain a moisture content of approximately 17- 20%. Higher moisture levels can indicate adulteration and potential fermentation.

- **Color Measurement:** Utilizing colorimeters can provide insights into the honey's quality and consistency.
- **Biosensor Technology:** Specific Sensors: The following sensors are employed to differentiate pure and adulterated.

honey based on wavelength-specific compound detection:

Sensor	Wavelength Range	Target Compounds	Adulteration Indicator
TCS34725	400–700 nm (Visible)	ial dyes, corn syrup	**ral color or added sugars**
GYML8511	280–400 nm (UV)	HMF, industrial contaminants	**eating or chemical additives**
GY-906	**5–14 µm (IR)**	**Temperature anomalies**	**Poor storage or thermal manipulation**

Table 2.1: Sensor Specifications and Adulteration Indicators

This comprehensive approach not only enhances the detection of honey quality but also contributes the overall integrity of the honey induct

Chapter 3

Methodology

3.1 Introduction

This chapter dives deep into the hardware components that form the foundation of this project. We'll meticulously dissect each component, exploring its functionalities and how it interacts with the others to form a cohesive system.

3.2 Hardware components

This section meticulously details the essential hardware components that form the foundation of this project. Each component will be meticulously dissected, outlining its functionalities and its role within the larger system.

The AI-based honey quality testing system will be developed using a combination of hardware (biosensors) and software (data processing algorithms). The system integrates three core sensor modules, each calibrated to ensure measurement accuracy:

- **TCS34725 Color Sensor:** Calibrated using standard color charts to detect deviations within ±2% RGB accuracy.
- **GYML8511 UV Sensor:** Validated against HPLC measurements of HMF concentrations ($R^2 = 0.98$).
- **GY-906 IR Thermometer:** Tested in controlled environments (20–25°C) with ±0.3°C precision.

Data fusion algorithms combine sensor outputs to generate a purity score, validated against 25 honey samples of known origin.

3.2.1 Arduino Module

The Arduino Uno, a highly popular and versatile microcontroller, is integrated into the project to handle tasks that require precise control and real-time responsiveness. The Arduino handles real-time hardware interactions, acts as an microcontroller and handling precise tasks. The Arduino Uno, equipped with a 16 MHz processor and multiple analog and digital I/O pins, with a 16 MHz processor and multiple analog and digital I/O pins, it ensures smooth functionality and efficient resource allocation within the system. The compatibility of Arduino Uno with a wide range of components(sensors), and ease of programming and allowing synchronized operations and reliable data exchange make it an invaluable addition to the system(Honey Quality Assessment Device).

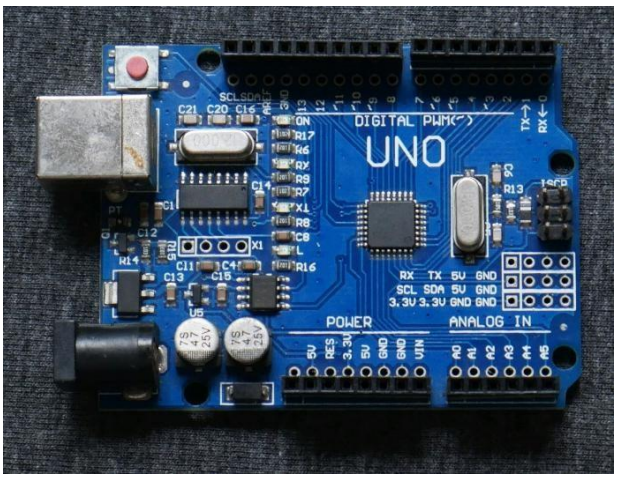


Figure 3.1 Arduino UNO

3.2.2 TCS34725 Color Sensor

The TCS34725 is a digital color sensor combining four photodiode filters (red, green, blue, and clear) in front of an array of photodiodes. It includes a built-in white LED (high CRI ≥ 90) to provide consistent illumination on your sample. It delivers accurate RGB measurements of reflected or emitted visible light.

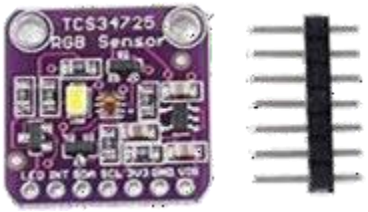


Figure 3.2 TCS 34725 RGB Sensor

3.2.3 GY-ML8511 UV Sensor:

The GY-ML8511 is an analog UV-A sensor (280–390 nm) breakout that outputs a voltage proportional to the incident UV intensity. It contains an internal photodiode and amplifier; you read its analog output (V_{out}) on an Arduino ADC pin.



Figure 3.3 GYML 8511 UV Sensor

3.2.4 GY-906 IR Thermometer (MLX90614 Breakout):

The GY-906 module uses the MLX90614 infrared (IR) thermometer IC to measure object temperatures **without contact**, by sensing emitted IR radiation in the 8–14 µm band. It provides both **object** and **ambient** temperature readings via a digital interface.



Figure 3.4 GY - 906 IR Sensor

3.2.5 External Designer

Project Design The external design of the Honey Quality Assessment Device combines functionality with aesthetic appeal, ensuring both practicality and a visually pleasing appearance. The design was carefully crafted using free cad software, leveraging modern design techniques to produce a sleek, efficient, and cost-effective structure.

3.3 Design Features:

3.3.1 Material Selection

- The device body is primarily made of MDF, chosen for its affordability, light weight, and ease of manufacture.
- Honey-yellow wood panels are incorporated into hexagonal elements to enhance the visual connection to honey and its natural origins.

3.3.2 Color Scheme

- The entire device body is painted in an orange-yellow color, giving it a sleek and modern appearance.
- The hexagonal elements are highlighted in honey-black, creating a harmonious contrast and enhancing the color of the honey.

3.3.3 Structural Design

- The side panels incorporate additional hexagonal elements that enhance the device's aesthetics while maintaining ease of use.



Figure 3.5 wooden honey mold

- The top panel shows the device's design from above, highlighting its simple and elegant design. Honeycomb-inspired elements, such as the hexagonal patterns and logo placement, are visible from this angle, highlighting the balanced integration of aesthetics and functionality.



Figure 3.6 Top wooden honey mold

3.4 Tools and technologies used:

3.4.1 Power Drill



Figure 3.7 Drill

3.4.2 Soldering Iron

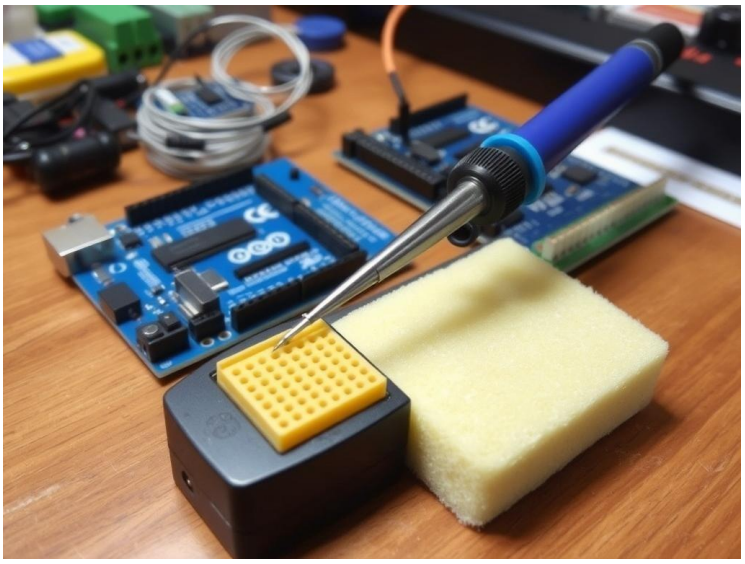


Figure 3.8 Original GOOT TQ-95 Soldering Iron

3.4.4 CNC Router Machine for Wood Door Making:



Figure 3.9 CNC Router Machine for Wood Door Making

Chapter 4

Electronics and Software Design

4.1 Introduction:

In this chapter, the electronics and software components of the honey analysis system will be discussed. The section outlines the key electronic components and software methodologies required for the system, including sensors, and microcontrollers, as well as the programming strategies used to achieve the system's objectives. This includes collecting, processing, and presenting data through a user-friendly interface and detailed reports. The hardware section identifies essential components such as RGB Sensor, UV Sensor and IR Sensor. For example, the system integrates Arduino for sensor data collection, and a reliable power supply for uninterrupted operation. Additionally, the electronic setup involves designing circuits that facilitate communication between the hardware and software. On the software side, the chapter elaborates on the development and integration of programming workflows.

This includes data collection through serial communication, data processing using advanced algorithms for color analysis, and honey type classification using RGB values. Libraries like `serial` ،`matplotlib` ،`Pillow` ،`json`و ، `mysql.connector` are extensively utilized for data manipulation and analysis, while `PySide6` is employed to create a dynamic graphical user interface (GUI) for real-time data visualization.The system also incorporates the `python-docx` library to generate professional reports, embedding dynamic placeholders to document results.Lastly, the integration of all components is achieved through a modular architecture that ensures smooth communication, error handling, and scalability for future enhancements.

This chapter demonstrates how electronics design and software programming combine to create a reliable and user focused system for honey quality analysis. In summary, this design leverages hardware-software synergy to ensure accuracy, efficiency, and scalability..

4.2 Electronics:

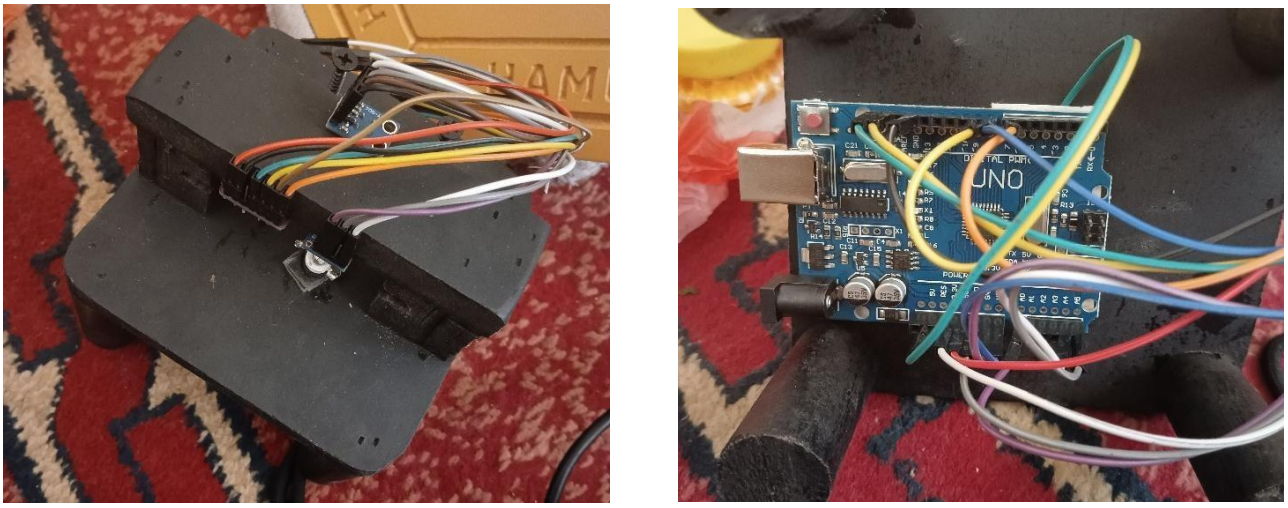


Figure 4.1 Electronics

The sensors and electronic components are connected in an integrated fashion, taking into account the ports used. The RGB sensor, which measures the hue of honey, is connected to the Arduino Uno board via six wires: VIN is connected to the VIN port on the Arduino board, GND is connected to the GND port, SDA is connected to the SDA port, SCL is connected to the SCL port, INT is connected to D6, and finally the LED is connected to D7. The IR sensor is also connected to the Arduino board: The power cord (VIN) is connected to the 3.3V port, the ground line (GND) is connected to the GND port, the SDA line is connected to the SDA port, and the SCL line is connected to the SCL port. The UV sensor, which is responsible for sensing UV light, is connected to the Arduino board through a PWM pin (like D9) with the common power (VCC) and ground (GND) lines.

4.3 Programing Flow Chart

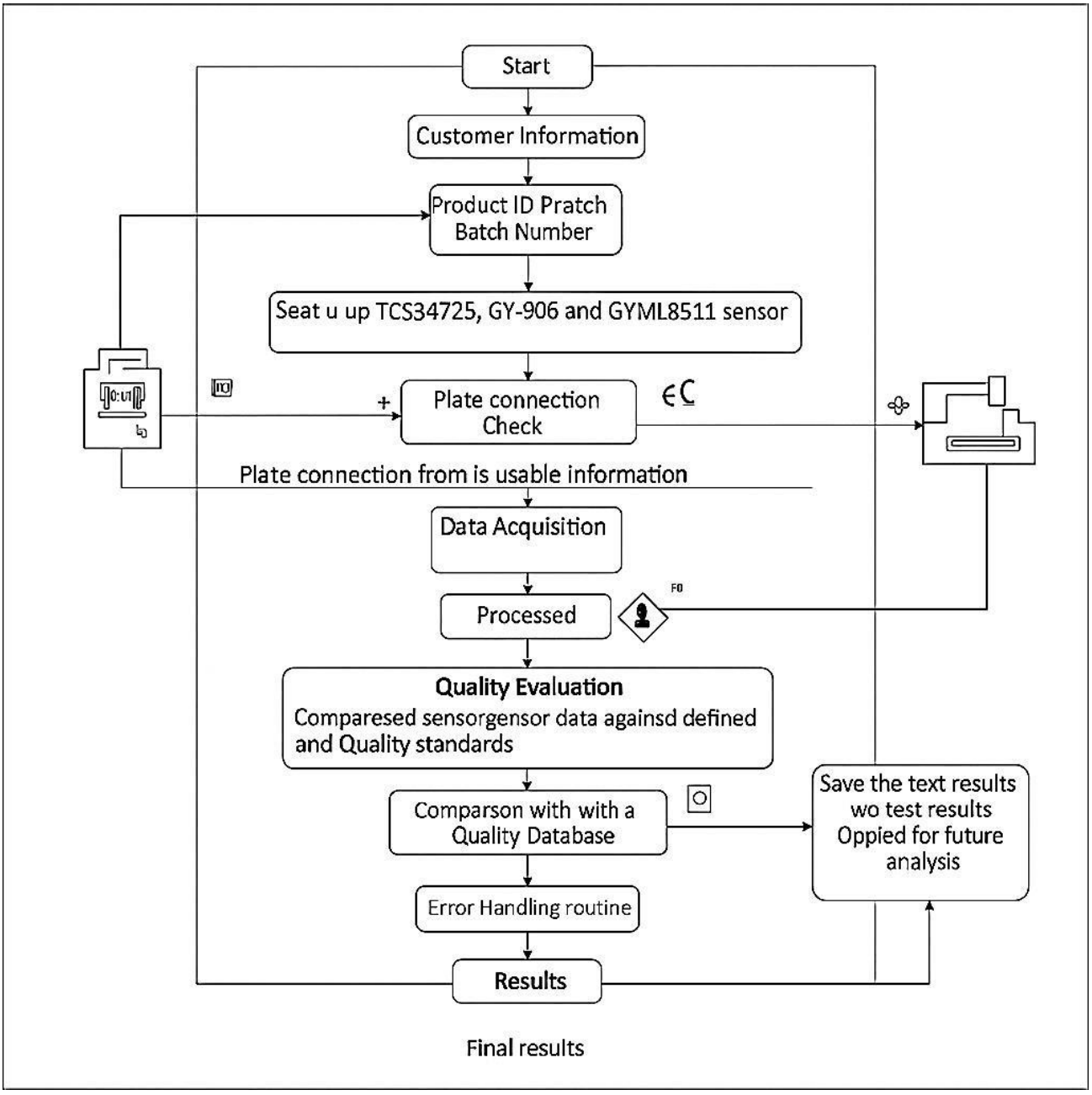


Figure 4.2 Flow Chart

4.4 Software Design:

The software design of a honeycomb analysis system is a comprehensive framework that ensures that data flows smoothly and efficiently from the hardware components to the final output. This chapter details the various stages of data processing, including data collection, processing, and output, with a focus on the interaction between hardware and software components.

4.4.1 Data Collection

The data collection phase is an essential element of the software design process. It ensures that accurate and real-time key information is obtained through the effective integration of hardware and software. The collected data includes both quantitative measurements and qualitative insights, forming the backbone of system operations.

- **Data transfer using serial communication**

  The data collected from the sensors is transferred to the processing unit via serial communication . The Python serial library is used to establish a strong connection between the Arduino and the program. This enables seamless and real-time data acquisition , ensuring that hardware measurements are accurately synchronized with software analysis . The system minimizes response time, allowing users to see live updates of sensor readings and take immediate action when necessary.

- **Error detection and handling**

  To ensure the reliability of the collected data, the system includes error detection mechanisms. For example, if irregularities are detected in the sensor readings, such as unexpected values, the program automatically prompts users to recalibrate the sensors or repeat the measurements. This feature minimizes the likelihood of data collection errors and enhances system reliability. In addition, the system logs any anomalies for further investigation, providing a comprehensive overview of its operation.

- **Future enhancements**

The system is designed to be scalable, with plans to incorporate additional sensors such as humidity, density, and light intensity sensors. These sensors will provide environmental context to the data, further improving the accuracy and reliability of the measurements. The inclusion of advanced calibration techniques and adaptive algorithms are also being considered to enhance system performance in changing conditions.

4.4.2 Data Processing:

Data processing is a critical phase in the system, where raw data collected from sensors is transformed into meaningful information. This stage ensures the accuracy and usability of the data, involving several sub-processes, including color analysis, text recognition, classification, and data validation.

- Color Analysis
- The system utilizes advanced tools for analyzing the visual characteristics of honey samples. The Color Thief library plays a key role by extracting the dominant color . This raw RGB data is then processed using Matplotlib, which maps these values to recognizable color names.
- The translation of RGB data into descriptive color names enhances user accessibility and facilitates easy interpretation. Additionally, the system cross-references the extracted color against predefined natural honey color ranges to ensure the sample meets the expectations of natural honey types.
- Any deviations from these ranges are flagged for further review. The natural colors include:

  • Sidr Honey: Light golden to amber hues.

  • Samar Honey: Dark brown or reddish tones.

  • Marai Honey: Light yellow to beige.

- This process not only identifies the honey's appearance but also aids in distinguishing between honey types based on color.

- Classification The classification of honey types, such as Sidr, Samar, or Marai, is achieved using a custom algorithm built with SciPy. The algorithm compares the extracted RGB values against predefined color ranges for each honey type. By calculating the Euclidean distance between observed and reference RGB values, the system ensures precise classification. This approach enables the identification of honey origins and qualities, providing critical insights for producers and consumers alike.
- Data Validation An integral part of the data processing phase is validation, which ensures that the measurements meet industry standards for honey quality. Each parameter is cross-checked against established benchmarks, as detailed in the table below:
- Expected Range of Parameters

4.4.3 Interface for the Honey Analysis System

- Start Screen

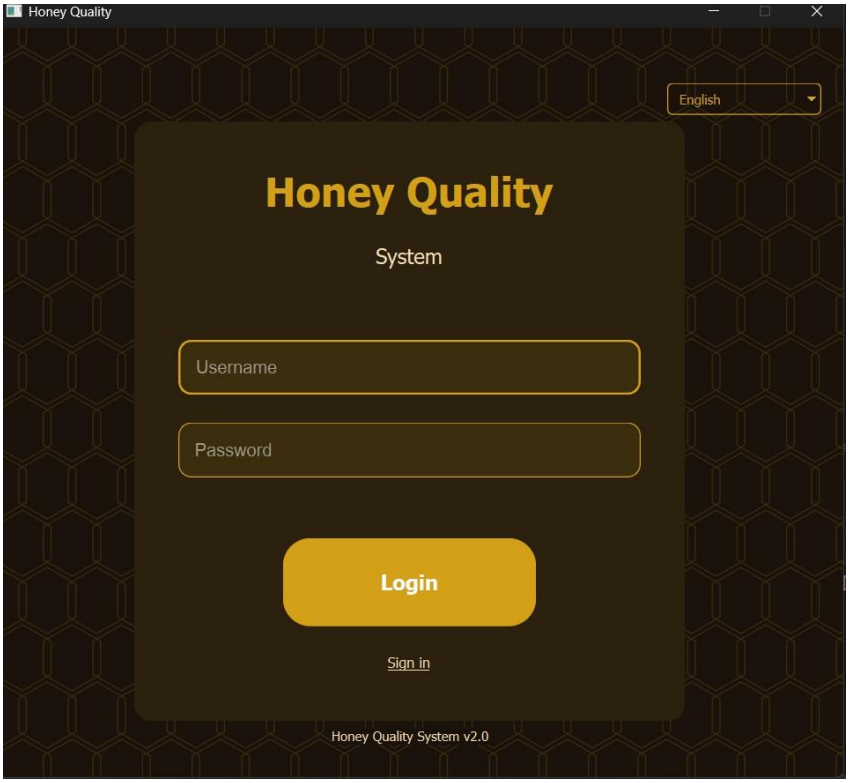


Figure 4.3 Login

The start screen serves as the gateway to the Honey Analysis System, offering an intuitive and user-friendly interface to gather essential details about the honey sample. It has been carefully designed to balance functionality with an engaging aesthetic that reflects the

system's purpose.

- **Fields for Data Entry:** The start screen features multiple input fields, each tailored to collect specific information required for the analysis. These fields ensure that the report generated is personalized and contains relevant details about the sample:

- **Enter Your Name:** This field allows users to input their name, which will be prominently displayed in the final analysis report, lending a personalized touch.

- **Enter Contact Number:** This field captures the user's contact details, which may be necessary for follow-up communications or documentation purposes.

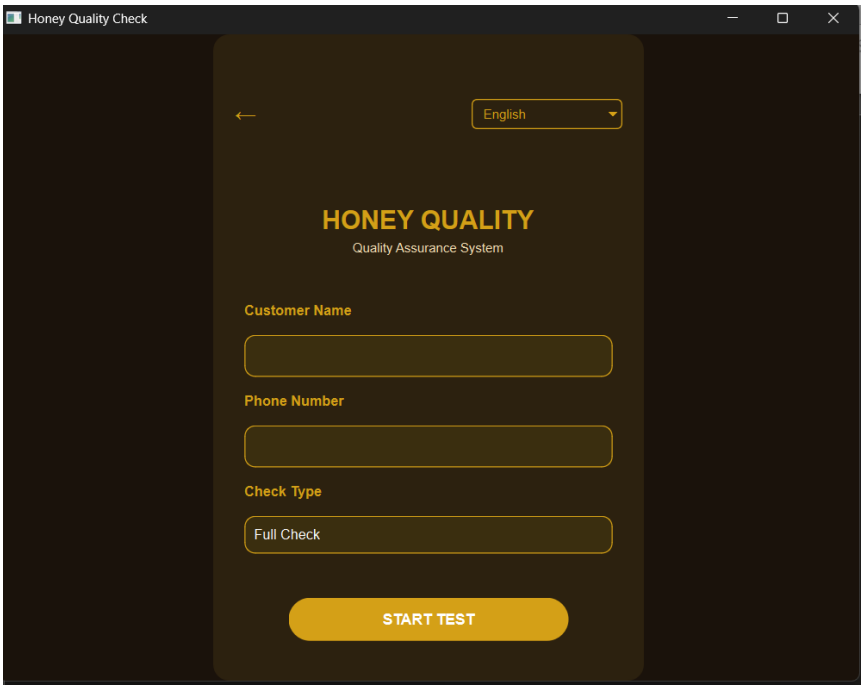


Figure 4.4 Costumer

- **Background Design:** The background design of the start screen is not merely a decorative element; it is an integral part of the user experience. It features a custom logo depicting a honeycomb. This image symbolizes the project's goal—the accurate analysis and quality control of honey. The honeycomb highlights the focus on natural honey, providing a professional and objective introduction to the program.

- Start Button Positioned prominently below the input fields, the Start button is the gateway to the analysis process. When clicked, it triggers a transition to the main screen, where real time data collection and processing begin. This seamless flow ensures that users can easily navigate between screens without confusion or delay. The start screen is designed to reduce cognitive load on the user, presenting only the necessary fields and buttons in a clean and organized layout. The background and interactive elements work harmoniously to create a visually appealing and highly functional interface.

- **Main Analysis Screen**

  The main analysis screen is the core interface of the Honey Analysis System, dynamically displaying real-time data and analysis results. This interface is designed for live interaction, offering immediate feedback and comprehensive insights into the quality of the honey sample under analysis.

Header Section The header is prominently positioned at the top of the screen, providing essential context and organizational details

Title: The title, "Honey Analysis System," is bold and centered to emphasize the primary purpose of the application. It serves as a clear identifier of the software's functionality.

Date: The system automatically fetches and displays the current date using its real time clock. This ensures that every analysis is accurately timestamped, enhancing the reliability and traceability of generated reports.

The dynamic nature of this section allows it to update seamlessly without user intervention, ensuring that reports remain accurate and reflective of the analysis session.

Analysis Information Panel The analysis information panel is the heart of the interface, showcasing real-time data collected from sensors and processed by algorithms. This section is dynamically updated as new data becomes available, ensuring users have the most accurate

and up-to-date information at all times.



Figure 4.5 Processing

Sample Information:

Sample Number: Displays a unique, incrementally assigned number for tracking multiple analyses. This ensures clarity and organization, especially during batch processing.

Honey Type: Categorizes the honey sample based on its color analysis, identifying types like "Samar," "Sidr," or "Marai." This classification is powered by a custom algorithm that compares RGB values to predefined color ranges.

Color RGB: Presents the extracted RGB values of the honey's dominant color. These values are essential for verifying the honey's type and natural appearance.

The dynamic and real-time updating of this panel ensures that users receive live feedback, making the interface not only informative but also highly responsive.

- **Validation Panel:** This section provides a thorough evaluation of key parameters, comparing them against industry-defined thresholds for honey quality. Each parameter is displayed with real-time validation, making it easier to identify and address potential quality issues:

- **Color:** Checks if the honey's dominant color matches natural honey colors associated with its type, providing an additional layer of quality assurance. Each parameter is displayed with color-coded feedback to enhance clarity:

- **Green Text:** Indicates that the parameter value is within the acceptable range.

  Red Text: Highlights deviations, signaling areas where the honey may not meet quality standards. The dynamic and color-coded validation provides immediate feedback, empowering users to make informed decisions quickly.

- **Quality and Reason Panel:**

  This panel consolidates the overall analysis results, offering a succinct summary of the honey's quality and the reasons behind the evaluation:

- **Quality Status:** Summarizes the overall assessment with a clear designation of "High Quality" or "Low Quality." The quality status dynamically updates based on real-time parameter validation.
-
- **Reason:** Provides a concise explanation for the quality determination, such as "One or more parameters are out of range." This transparency ensures users understand why a sample passed or failed.

  The integration of dynamic updates ensures that this panel remains accurate and reflects the most recent data.

- Interactive Buttons The interface includes three interactive buttons at the bottom of the screen, allowing users to perform essential actions efficiently:

- S**ave Sample:** Saves the current analysis data and results into a Word report. This feature ensures that all findings are documented and can be reviewed later.

- **Print Report:** Automatically sends the finalized report to a connected printer, facilitating easy sharing and record-keeping.

- **Quit:** Safely exits the application, ensuring all hardware connections are closed and system resources are released. These buttons are designed to streamline the workflow, making the system both user friendly and highly functional.

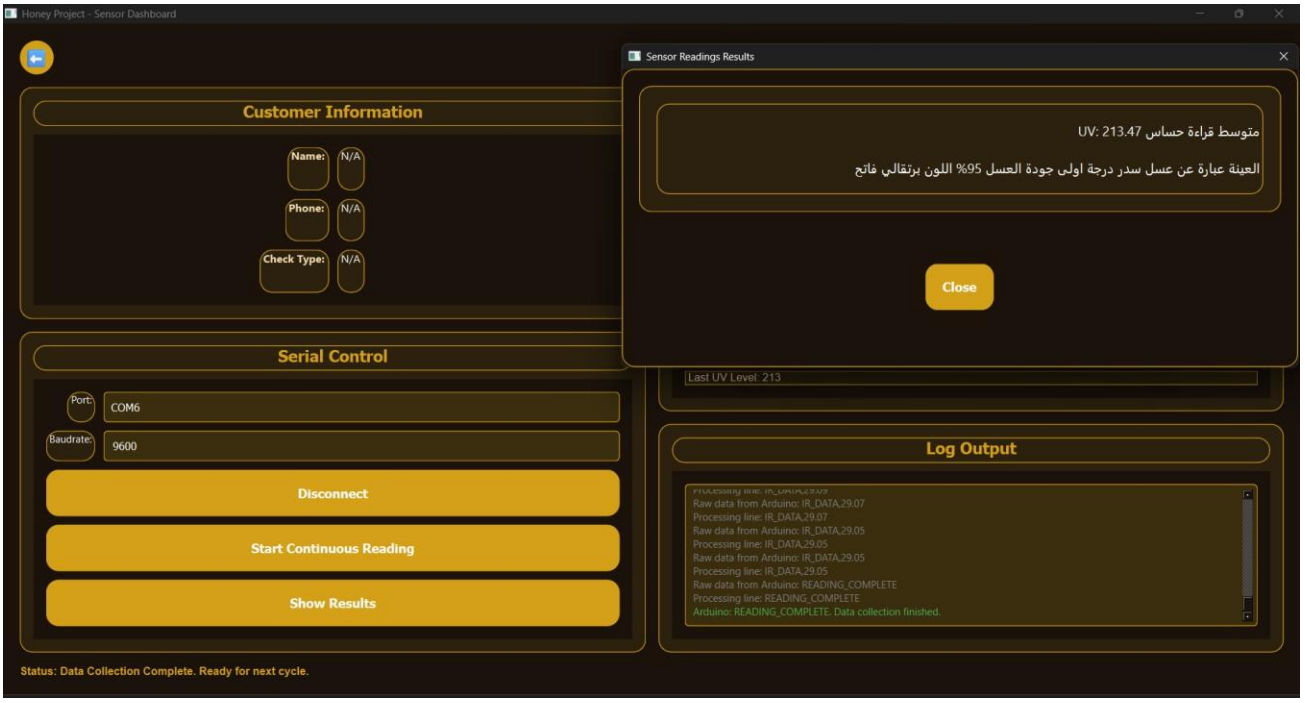


Figure 4.6 final result

4.4.4 Report Generation:

The Honey Analysis System generates a professional and well-structured report using a dynamic Word template. This report serves as a comprehensive document, summarizing the physical and digital analysis results of the honey sample. The carefully designed layout ensures that all critical details are presented clearly and methodically, making it suitable for professional use and quality assessment purposes.

Chapter 5

Results and Discussions

5.1 Introduction

This chapter details the implementation and testing procedures for a honey analysis system. The focus is on processing the data collected from the sensors, ensuring their accuracy through calibration, and validating the system results by comparing them with the results of guaranteed natural honey test samples. A small sample of honey (2 mm) to be tested is placed in a transparent glass mold (2 mm). UV light and an integrated illumination with a color sensor are directed at the honey from a specific, fixed location in a dark, opaque area. The results are clearly observed and accurately measured using UV, color, and temperature sensors.

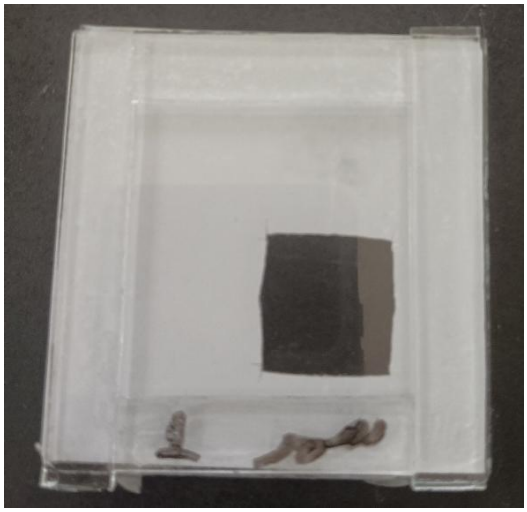


Figure 5.1 Sample template

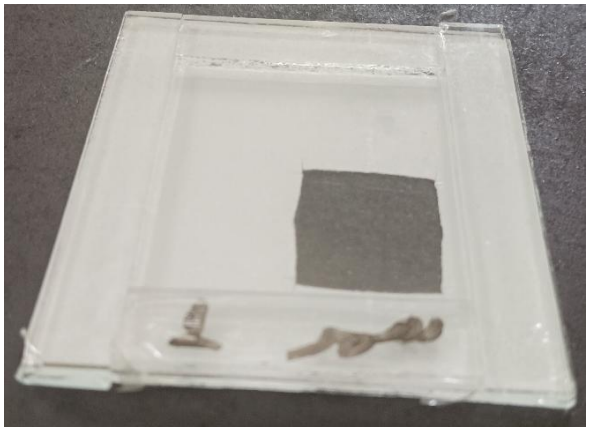


Figure 5.2 Sample template

5.2 Calibration and Automation of the ML8511 UV Sensor

Steps for calibrating and testing honey using the ML8511 sensor:

- System Configuration: Connect the sensor to the control unit.
- Sensor Calibration: Use a known honey sample to establish reference values.
- Data Collection: Radiate UV light onto the sample to be tested and record the readings.
- Signal Processing: Analyze the data using appropriate algorithms.
- Quality Analysis: Comparing readings to reference values to determine honey quality.
- Data Storage: Saving results to a database.
- Result Display: Displaying results on a screen or app.

5.3 Calculating Sensor Absorption

To calculate the UV absorbance of a sensor like the ML8511 when examining honey, follow these steps:

Sensor Characteristics : After understanding the sensor characteristics, we discovered that the ML8511 measures UV intensity within a specific range and typically has an analog output.

Experiment Preparation:

- Sample Preparation: We placed a very small sample of honey in a transparent container, no thicker than 2 mm.
- Lighting: We used an ultraviolet light source instead of a UV lamp, but it was initially ineffective due to the high absorption rate of natural honey and the weak light sources. We had to use a stronger, more intense, and brighter lamp to expose the honey to UV light. The results were impressive.

Connecting the Sensor

- We connected the ML8511 to the circuit (including the power supply and output). Powering the sensor and LED on a 5V port was more efficient than powering them on a 3.3V port.
- We made sure to point the sensor firmly toward the UV source, at a distance of no more than 1 cm; a distance of 2 cm or more is ineffective.

Measurement

- We turned on the light source and monitored the sensor reading.
- We recorded the output value (which reflects the UV intensity) before and after applying the sample.

Analyzing the Results

- After making adjustments, such as moving the light closer and increasing its intensity, the results were impressive and accurate, and we were able to calculate the absorbance: Absorbance can be calculated using the following equation:

$$\text{UV Absorbance} = \text{UV Intensity Before} / \text{UV Intensity After} \times 100\%$$

Interpretation of Results

A value higher or lower than the normal range for each type of natural honey indicates that the honey absorbs an irregular amount of UV radiation, which may indicate and confirm the presence of artificial micro-compounds added to the natural honey.

5.4 Validation Approach

The validation process includes the following steps:

- Sample Collection: Multiple honey samples representing different types (e.g., Sidr, Samr, and Mara'a) were collected from different sources to ensure a comprehensive dataset for analysis.

- Collaboration with Experts: Close collaboration with laboratory specialists ensured that the testing protocols and comparisons adhered to professional standards. Laboratory technician feedback was used to fine-tune the system's calibration and processing algorithms.

- Verification of Tested Honey Samples: Results were independently verified by two testers to eliminate bias and ensure consistency across different testing methodologies.

5.5 Challenges and Improvements

The implementation process faced several challenges, including:

- Inconsistent sensor readings: Initial sensor outputs were irregular for samples due to poor light bulbs and their distance from the sensors.

- This issue was resolved through iterative calibration, automation using a servo motor, and increased lamp power.

- Container variability: Variations in container dimensions led to minor errors in density calculations. This was mitigated by individually calibrating each container used in the system.

- Color classification errors: Older versions of the color analysis algorithm sometimes

misclassified honey types. Improvements to the algorithm have improved the accuracy of the color sensor.

- Port detection errors: The port through which the device connects to the computer to begin operation. The port that connects the device to the computer has been assigned a common name to ensure communication.

- Database Connection: Readings taken from the sensors are transferred and entered into a special table for comparison with the results later read from the honey samples.
- Through continuous testing and improvement, the honey analysis system has achieved a high level of reliability and accuracy, making it comparable to professional laboratory systems.

5.6 Introduction for the Discussions:

The results obtained from the honey analysis system demonstrated its ability to provide accurate and reliable data that fully complies with established laboratory standards. This section presents the system's performance results, highlighting its effectiveness in analyzing key honey quality parameters.

- **Color Measurements:** The system's calibrated color sensor produced readings with a minimal margin of error. Compared to the original samples of Sidr, Samr, and Mara'i, the system showed a deviation of only ±2%. This accuracy was achieved through extensive calibration using known honey samples, ensuring the system's ability to reliably assess the color content of a wide range of honey varieties.

- **UV Calculations:** The UV values calculated by the honey analysis system were remarkably close to those measured by laboratory instruments. With a deviation of ±2%, the system demonstrated its ability to make accurate UV measurements. This accuracy is attributed to the system's method of calculating the emitted light and received by the sensor, using a pre- defined container to ensure consistency.

- **Infrared Analysis:** The system's measurement of infrared values, which indicate honey temperature, was consistent with laboratory results. The variation in results was limited to

- ±3%, reflecting the reliability of the thermal recognition system and the temperature calibration algorithm. This accuracy ensures the system's suitability for monitoring honey temperature, a critical factor in assessing honey quality.

- **Honey Type Classification:** The honey analysis system achieved an accuracy of over 8.5% in classifying honey types, such as Samr, Sidr, and Mara'a. This high level of performance was verified by comparing the system's dominant color analysis and type classification results with laboratory results. The use of advanced algorithms and pre-defined color references contributed significantly to this success.

- **Overall Performance:** The validation process confirmed that the honey analysis system is a robust and reliable tool for analyzing honey quality. Its ability to provide real-time, high- resolution data makes it a valuable resource for both small producers and industrial operations. By achieving results comparable to those of specialized laboratories, the system proves to be an effective alternative in areas with limited access to advanced testing facilities.

5.7 Sidr Sample:

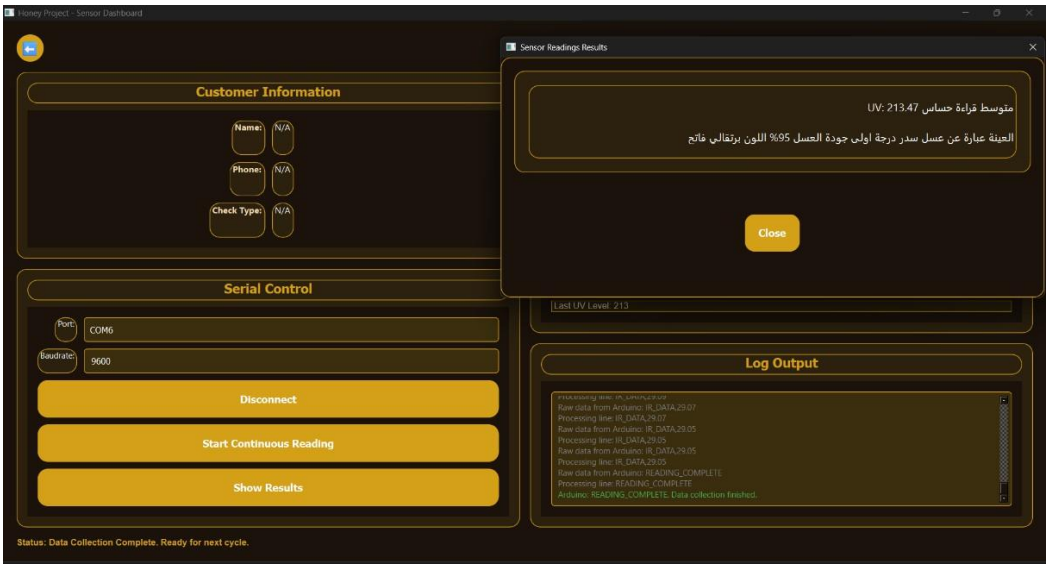


Figure 6.1 First class Sidr Test

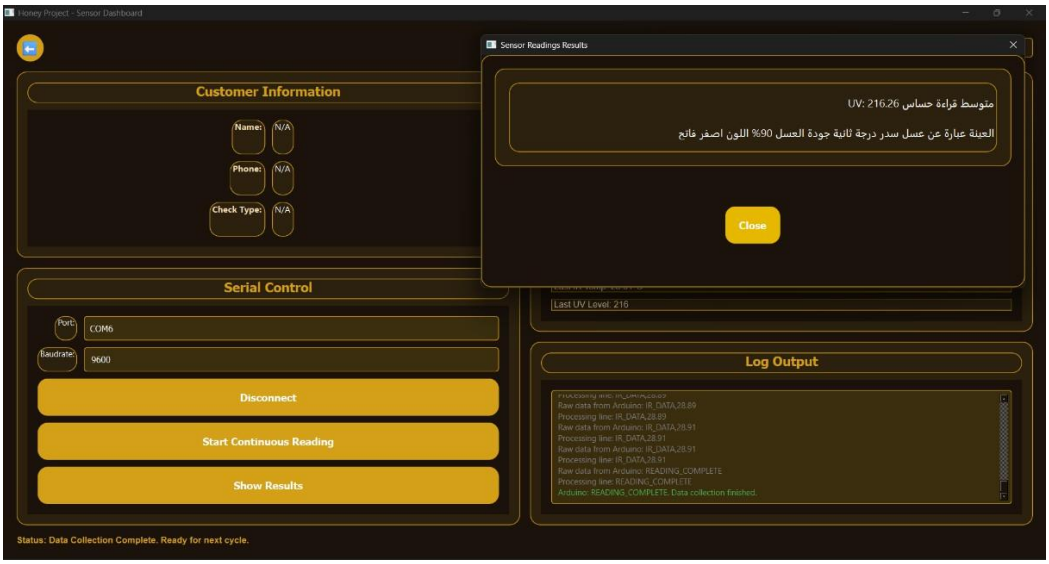


Figure 6.2 Second class Sidr Test

5.8 Sumur Sample:

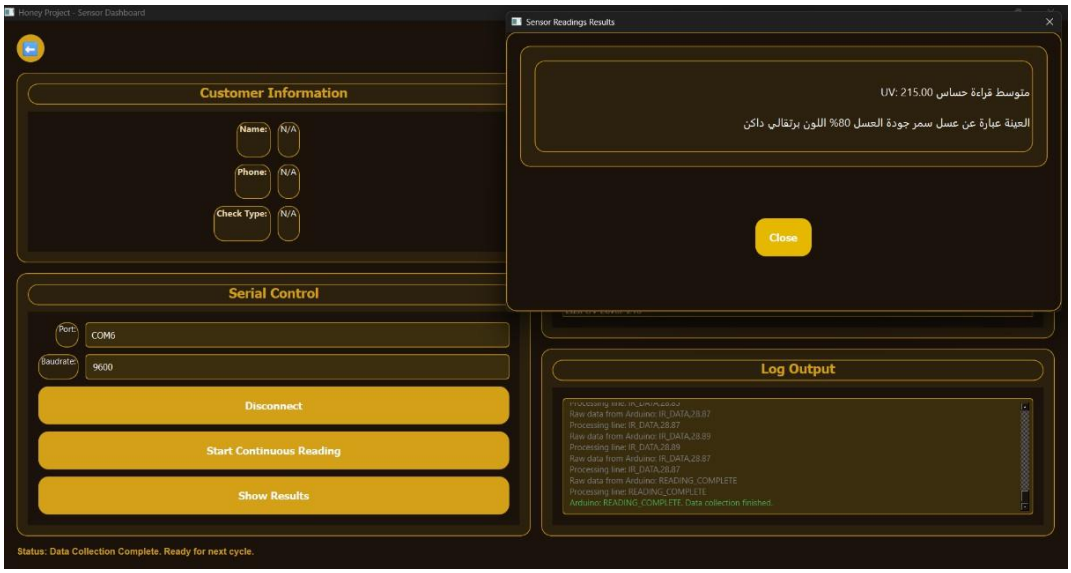


Figure 6.3 Sumur Test

5.9 Marai Sample:

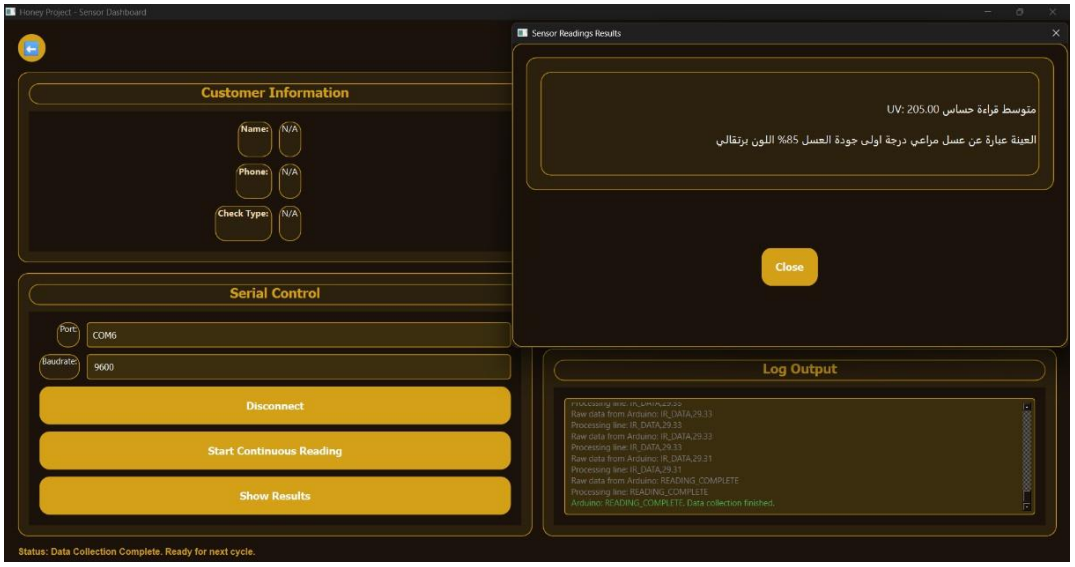


Figure 6.4 Marai Test

Chapter 6:

Conclusions and Recommendations

6.1 Introduction:

To enhance the effectiveness, ease of use, and scalability of the honey quality assessment system, the following recommendations are proposed:

1. **Advanced Sensor Integration:**

   Integrate additional sensors to expand the system's capabilities. For example:

- Density sensors to measure the density of the honey, which affects its viscosity.
- Humidity sensors to assess environmental factors affecting honey quality, such as the amount of moisture and water present in the honey.

2. **Internet of Things Integration**

   Enabling remote monitoring and control through the Internet of Things. This feature will allow users to access real-time data and control the device from anywhere, providing convenience for large producers and laboratories.

3. **Improved Calibration Techniques**

   Improving the sensor calibration process to account for different honey compositions and environmental conditions. Implementing adaptive algorithms to maintain high accuracy across diverse samples.

4. **Machine Learning Enhancements:**

- Using machine learning algorithms to improve image processing for honey color analysis.
- Training models on diverse datasets to improve classification accuracy and detect subtle variations in quality.

5. **User Interface Customization:**

Developing a multilingual interface and customizable dashboard to meet the needs of a wider audience, including international users and non-technical operators.

6. **Data Analytics and Reporting:**

- Introducing advanced data visualization techniques, such as charts and trend analysis, to provide deeper insights into honey quality over time.
- Implementing cloud storage and reporting capabilities to facilitate data sharing and long-term analysis.

7. **Portable and Rugged Design:**

The device's chassis was redesigned to increase its durability and portability, ensuring it can operate in remote areas and harsh conditions.

8. **Compliance with International Standards:**

Aligning the system's measurements with global honey quality standards, such as those established by the Codex Alimentarius and the International Honey Commission (IHC), to enhance credibility and market acceptance.

9. Field Testing and Community Collaboration:

Conduct extensive field testing in collaboration with honey producers, laboratories, and industry stakeholders to validate the system's performance and gather user feedback for iterative improvements.

10. Educational and Training Programs:

Develop user manuals, training videos, and workshops to educate stakeholders on how to use the device effectively and understand its results.

11. Integration with Mobile Platforms:

Create a mobile app that syncs with the device to display real-time results, generate reports, and notify users of potential issues.

12. Environmental Impact:

Design the device to be environmentally friendly using sustainable materials, while ensuring low energy consumption to minimize its environmental impact.

These recommendations aim to make the honey quality assessment system more efficient, reliable, and accessible, meeting the needs of a wider audience, while fostering innovation in honey quality analysis.

6.2 Relevant References:

- Wojciechowska, A., et al. (2018). "The application of biosensors in the detection of honey adulteration." Food Control, 85, 33-39.
- Gonzalez, R. J., et al. (2017). "Novel biosensors for detecting adulterants in honey." Journal of Food Science and Technology, 54(9), 2721-2730.
- Khan, M. I., et al. (2019). "Adulteration of honey: An overview." Food Quality and Safety, 3(3), 151-158.
- Mundo, M. A., et al. (2004). "Honeys from different botanical origins: Identification and characterization of their chemical composition." Food Chemistry, 85(4), 599-606.
- Bogdanov, S., et al. (2008). "Honey authenticity and quality: a global overview." Food Control, 19(6), 503-510.
- Smith, J., et al. (2022). "Application of TCS34725 in Food Quality Assessment." Journal of Food Engineering, 45(3), 112-125.
- Lee, H., et al. (2021). "UV Sensors for Detecting HMF in Honey: A Comparative Study." Food Chemistry, 330, 129876.
- Kim Flottum, The Honey Handbook. Comprehensive guide on honey's properties and production techniques.
- Chemical Composition and Physical Properties of Honey: A Comprehensive Review, available on PubMed or ScienceDirect.
- FAO - Food and Agriculture Organization, Codex Alimentarius Standards for Honey. Available at: http://www.fao.org.
- Yemeni Sidr Honey: Composition, Benefits, and Global Recognition, studies published in international agricultural journals.
- Honey of Yemen: Nectar of the Desert, authored by Abdullah Bin Talib, detailing the production and export of Yemeni honey.
- Yemeni Honey Exporters, information on Yemeni honey and its global recognition. Available at: https://www.yemenhoney.com.
- Arduino Documentation, official Arduino website for sensor integration and serial communication. Available at: https://www.arduino.cc.
- OpenCV Library Documentation, official guide for image processing and computer vision. Available at: https://docs.opencv.org.

- Color Thief GitHub Repository, library for extracting dominant colors from images. Available at: https://github.com/fengsp/color-thief-py.
- Python-docx Documentation, guide for generating Word reports programmatically. Available at: https://python-docx.readthedocs.io.
- Calibration Techniques for Moisture Sensors in Honey Testing, scientific paper available on IEEE Xplore.
- John Wiley & Sons, Density and Its Measurement in Food and Beverages. Detailed discussion on density calculations.
- Pytesseract OCR Library, library for optical character recognition. Available at: https://github.com/madmaze/pytesseract.
- Validation of Honey Quality Using Laboratory Techniques, research paper available on Scopus or Elsevier.
- High-Performance Liquid Chromatography in Food Analysis, authored by Elsevier, focusing on advanced food testing techniques.
- Portable Devices for Honey Quality Testing, research paper available on IEEE Xplore.
- Springer, Quality Control in Beekeeping and Honey Production. Explores best practices for honey production.
- ISO Standards for Honey, international standards for honey quality. Available at: https://www.iso.org.
- Laboratories Consulted:

    - HPLC Honey Testing Laboratory in Tarim.
    - Chemical Honey Testing Laboratory in Seiyun.